Chapter 10. Linear Regression

Least squares, permutation and bootstrap inference, robust regression and M-estimation

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10.1 Simple linear regression

Motivation. Rank-based slopes such as the median pairwise slope are robust to outliers and give distribution-free intervals.

Model $y_i = \alpha + \beta x_i + \epsilon_i$ with uncorrelated errors of mean 0 and variance σ^2; inference is conditional on the x_i. Least squares gives $\hat\beta = S_{xy}/S_{xx}$, $\hat\alpha = \bar y - \hat\beta\bar x$ and $s^2 = \sum e_i^2/(n-2)$.

Proposition 10.1

$E\hat\beta = \beta$ and $\operatorname{Var}\hat\beta = \sigma^2/S_{xx}$. The residuals satisfy $\operatorname{Var}(e_i) = \sigma^2(1-h_i)$ with leverage $h_i = \frac{1}{n} + \frac{(x_i-\bar x)^2}{S_{xx}}$, so the adjusted residuals $r_i = e_i/\sqrt{1-h_i}$ have variance σ^2.

Proof. $\hat\beta = \sum w_i y_i$ with $w_i = (x_i-\bar x)/S_{xx}$, $\sum w_i = 0$, $\sum w_i x_i = 1$, $\sum w_i^2 = 1/S_{xx}$. With the hat matrix H, $\mathbf{e} = (I-H)\boldsymbol\epsilon$ and $\operatorname{Var}\mathbf{e} = \sigma^2(I-H)$, whose diagonal entries are $\sigma^2(1-h_i)$. ■

Under normal errors $(\hat\beta-\beta)/\widehat{\mathrm{se}}(\hat\beta) \sim t_{n-2}$, $\widehat{\mathrm{se}}(\hat\beta) = s/\sqrt{S_{xx}}$; otherwise resample.

Permutation test of $H_0 : \beta = 0$. Under H_0 the $y_i = \alpha + \epsilon_i$ are exchangeable: shuffle y against fixed x. $\hat\beta$, $\sum x_i y_i$ and $t = \hat\beta\sqrt{(n-2)S_{xx}}/\sqrt{S_{yy} - \hat\beta^2 S_{xx}}$ (increasing in $\hat\beta$) give the same test, the permutation test of correlation (Section 5.1).

Bootstrap intervals. Case bootstrap (random x, robust to model failure) or residual bootstrap $y_i^* = \hat\alpha + \hat\beta x_i + r_i^*$ (fixed design). Since $\widehat{\mathrm{se}}(\hat\beta)$ has a formula, the bootstrap-t interval

$$\beta \in \big(\hat\beta - \widehat{\mathrm{se}}(\hat\beta)Q_{t^*}(1-\tfrac{\alpha}{2}),\ \hat\beta - \widehat{\mathrm{se}}(\hat\beta)Q_{t^*}(\tfrac{\alpha}{2})\big), \qquad t^* = \frac{\hat\beta^* - \hat\beta}{\widehat{\mathrm{se}}(\hat\beta^*)},$$

is cheap and second-order accurate. Tests of $\beta = 0$:

- null resampling: draw x and y *independently* (case form), or $y^* = \bar y + r_0^*$, $r_{0,i} = (y_i - \bar y)/\sqrt{1-1/n}$ (residual form);
- pivot resampling: compare t^* with $t_{\mathrm{obs}} = \hat\beta/\widehat{\mathrm{se}}(\hat\beta)$ (inverting the bootstrap-t interval).

10.2 Multiple regression

$\mathbf{y} = X\boldsymbol\beta + \boldsymbol\epsilon$, X of size $n \times (p+1)$: $\hat{\boldsymbol\beta} = (X'X)^{-1}X'\mathbf{y}$, $E\hat{\boldsymbol\beta} = \boldsymbol\beta$, $\operatorname{Var}\hat{\boldsymbol\beta} = \sigma^2(X'X)^{-1}$, $s^2 = \mathbf{e}'\mathbf{e}/(n-p-1)$ (unbiased: $E\,\boldsymbol\epsilon'(I-H)\boldsymbol\epsilon = \sigma^2\operatorname{tr}(I-H)$). Each β_j is a *partial* effect, adjusted for the other regressors.

Comparing nested models. Split $X = [X_0, X_1]$ ($k+1$ and $q = p-k$ columns) and test $H_0 : \boldsymbol\beta_1 = 0$. With hat matrices H and H_0,

$$F = \frac{\mathbf{y}'(H-H_0)\mathbf{y}/q}{\mathbf{y}'(I-H)\mathbf{y}/(n-p-1)} = \frac{\hat{\boldsymbol\beta}_1'\{X_1'(I-H_0)X_1\}\hat{\boldsymbol\beta}_1}{q\,s^2}. \tag{10.1}$$

Proposition 10.2

Under H_0 with normal errors, $F \sim F_{q,n-p-1}$. More generally $F_{\boldsymbol{\beta}_1}$, defined by replacing $\hat{\boldsymbol{\beta}}_1$ with $\hat{\boldsymbol{\beta}}_1 - \boldsymbol{\beta}_1$ in (10.1), has this distribution for every $\boldsymbol{\beta}_1$, so it is a pivot.

Proof. $H - H_0$ is a rank-q projection orthogonal to $I - H$, both annihilating X_0. Under H_0 the forms reduce to $\boldsymbol{\epsilon}'(H - H_0)\boldsymbol{\epsilon}$ and $\boldsymbol{\epsilon}'(I - H)\boldsymbol{\epsilon}$, independent $\sigma^2\chi^2_q$ and $\sigma^2\chi^2_{n-p-1}$ (Cochran). The second form is Frisch–Waugh–Lovell: $\hat{\boldsymbol{\beta}}_1$ regresses $(I - H_0)\mathbf{y}$ on $\tilde{X}_1 = (I - H_0)X_1$, and $H - H_0$ projects onto $\mathrm{col}(\tilde{X}_1)$. For general $\boldsymbol{\beta}_1$ use $\mathbf{y} - X_1\boldsymbol{\beta}_1$. ■

For $q = 1$, $F = t_j^2$; a one-sided $\beta_j > 0$ needs t_j, to which the schemes below apply equally. Bootstrap null resampling: the case form (draw $\mathbf{x}$-rows and y's independently; when all slopes are tested) or reduced-model residuals with replacement. Pivot resampling: case or full-model residual bootstrap with $F_{\hat{\boldsymbol{\beta}}_1}$ or $t_j^* = (\hat{\beta}_j^* - \hat{\beta}_j)/\widehat{\mathrm{se}}(\hat{\beta}_j^*)$; p-value $\#\{F^{*b} \geq F_{\mathrm{obs}}\}/(B+1)$ (observed value counted).

Permutation tests for $H_0 : \boldsymbol{\beta}_1 = 0$. Under H_0 the errors are exchangeable but the y_i are not (means depend on X_0), so no permutation test is exact. Approximate schemes:

1. *Shuffle y:* effectively tests $\beta_1 = \cdots = \beta_p = 0$; can be anticonservative.
2. *Freedman–Lane:* shuffle reduced-model residuals e_0, $y^* = \hat{y}_0 + e_0^*$, recompute F; generally the most accurate.
3. *ter Braak:* shuffle full-model residuals, $y^* = \hat{y} + e^*$, recompute the pivot $F_{\hat{\boldsymbol{\beta}}_1}$.

All are asymptotically equivalent; sampling with replacement in 2 and 3 gives the null and pivot residual bootstrap tests. Bootstrap-t intervals for β_j use t_j^* with $\widehat{\mathrm{se}}^2(\hat{\beta}_j) = s^2(X'X)^{-1}_{jj}$.

10.3 Robust regression

Least squares (the normal MLE) loses efficiency under heavy tails and can be ruined by one outlier.

Definition (Breakdown point)

The smallest fraction of observations that, sent to infinity, can send the estimator to infinity: $1/n \to 0$ for the mean, α for the α-trimmed mean, $1/2$ for the median.

Trimmed and Winsorized means. The α-trimmed mean drops the lowest and highest α fractions and averages the rest; the α-Winsorized mean replaces them by the α, $1-\alpha$ sample quantiles and averages all n. Both break down at α. Hodges–Lehmann breaks down at $1 - 1/\sqrt{2} \approx 29\%$: a fraction ε of wild points drags $1 - (1-\varepsilon)^2$ of the Walsh averages, exceeding $\frac{1}{2}$ when $\varepsilon > 1 - 1/\sqrt{2}$.

Example 10.1

For $-2, -1, -1, 0, 0, 1, 1, 2, 3, 50$: mean 5.3; 10% trimmed mean 0.625 (drop -2 and 50); 10% Winsorized mean 0.7 (replace them by -1 and 3); median 0.5. Replacing 50 by 10^{100} changes only the mean.

Trimming and Winsorizing treat both tails alike; M-estimators down-weight by residual.

10.3.1 M-estimators

Generalize $\bar{x} = \arg\min_\mu \sum (x_i - \mu)^2$ to

$$\hat{\mu}^\Psi = \arg\min_\mu \sum_i \Psi\Big(\frac{x_i - \mu}{\hat{\tau}}\Big) \iff \sum_i \psi\Big(\frac{x_i - \mu}{\hat{\tau}}\Big) = 0, \qquad \psi = \Psi',$$

with robust scale $\hat{\tau} = 1.4826$ MAD (consistent for σ under normality, $\Phi^{-1}(3/4) = 0.6745$).

- $\psi(x) = x$: the mean. $\psi(x) = \mathrm{sgn}(x)$: the median.
- **Huber:** $\psi(x) = \max(-k, \min(k, x))$; $k \approx 1.35$ gives about 95% normal efficiency.
- **Tukey biweight:** $\psi(x) = x(1 - (x/c)^2)^2 I(|x| \le c)$, redescending: gross outliers get weight zero.

A bounded ψ with a 50% breakdown scale breaks down at 50%. ("M" = maximum-likelihood type: $\Psi(x) = x^2/2$, $\hat{\tau} = \sigma$ gives the normal MLE.) Solve numerically, e.g. by IRLS with weights $\psi(u_i)/u_i$.

Inference on a location with outliers. Outliers inflate both the mean and s, and so drive the t-test and t-interval. Robust alternatives:

- sign test and order-statistic interval (Chapter 2), valid for the median under any continuous F;
- signed-rank test and Walsh-average interval, if the uncontaminated law is symmetric;
- an M-estimator with bootstrap SE and bias (normal, percentile or BCa interval); its asymptotic SE needs the influence function, so the bootstrap is practical.

Lower breakdown tends to mean wider intervals under contamination. In regression,

$$\hat{\boldsymbol{\beta}}^\Psi = \arg\min_{\boldsymbol{\beta}} \sum \Psi((y_i - \mathbf{x}_i'\boldsymbol{\beta})/\hat{\tau}),$$

and the equations $\sum \psi(\cdot)\,\mathbf{x}_i = 0$ show Huber is robust to outlying y but not to high leverage (x_{ij} unbounded). **Yohai's MM-estimator**: redescending ψ (biweight, $c = 4.685$ for 95% efficiency) from a high-breakdown S-estimate of scale, giving 50% breakdown and high efficiency. **Least trimmed squares** minimizes the sum of the h smallest squared residuals ($h = \lfloor n/2 \rfloor + \lfloor (p+1)/2 \rfloor$ for maximal breakdown, $h = \lfloor 0.9n \rfloor$ for 10% trimming); **least median of squares** the median squared residual. Both are very robust, inefficient, and used as starting values.

10.3.2 The Theil–Sen estimator

For $y_i = \alpha + \beta x_i + \epsilon_i$, with pairwise slopes $\beta_{ij} = (y_j - y_i)/(x_j - x_i)$ $(x_i \ne x_j)$,

$$\hat{\beta}^{TS} = \mathrm{median}_{i<j}\, \beta_{ij}, \qquad \hat{\alpha}^{TS} = \mathrm{median}_i (y_i - \hat{\beta}^{TS} x_i).$$

Proposition 10.3 (Link to Kendall's tau)

Let $\tau_n(\beta_0)$ be Kendall's statistic between x_i and $y_i - \beta_0 x_i$. Then $\binom{n}{2}\tau_n(\beta_0) = \#\{\beta_{ij} > \beta_0\} - \#\{\beta_{ij} < \beta_0\}$. Hence the test of $H_0 : \beta = \beta_0$ by Kendall's tau, and the interval $(\beta_{(l)}, \beta_{(u)})$ with $l = \binom{n}{2}\frac{Q_K(\alpha/2)+1}{2}$, $u = \binom{n}{2}\frac{Q_K(1-\alpha/2)+1}{2}$ (Q_K the null quantile of τ_n), are exact. $\hat{\beta}^{TS}$ is the corresponding point estimate.

Proof. $\mathrm{sgn}[(x_i - x_j)\{(y_i - \beta_0 x_i) - (y_j - \beta_0 x_j)\}] = \mathrm{sgn}[(x_i - x_j)^2(\beta_{ij} - \beta_0)] = \mathrm{sgn}(\beta_{ij} - \beta_0)$. At the true β, $y_i - \beta x_i = \alpha + \epsilon_i$ is independent of x_i, so $\tau_n(\beta)$ has the null law K_n. The acceptance region

is a range of $\#\{\beta_{ij} > \beta_0\}$, a step function of β_0 dropping by one at each ordered slope, hence an interval of ordered slopes with the stated indices. ■

$\hat{\beta}^{TS}$ breaks down at $1-1/\sqrt{2} \approx 29\%$ and has about 96% ARE vs. least squares under normal errors.

Example 10.2

Points $(1,1),(2,2),(3,3.5),(4,4),(5,20)$, the last an outlier. Ordered slopes $0.5, 1, 1, 1, 1.25, 1.5, 4.75, 6, 8.25, 16$ give $\hat{\beta}^{TS} = 1.375$, $\hat{\alpha}^{TS} = -0.625$; least squares gives 4.0.

In multiple regression one version uses **backfitting**:

1. Start with single-variable Theil–Sen slopes $\hat{\beta}_j^{(0)}$ and set $\hat{\beta}_0^{(0)} = \text{median}_i(y_i - \sum_j \hat{\beta}_j^{(0)} x_{ij})$.
2. For $k = 1, 2, \ldots$ and each j, take the Theil–Sen slope of $y_i - \sum_{l \neq j} \hat{\beta}_l^{(k-1)} x_{il}$ on x_{ij}; update the intercept as a median.
3. Iterate until the coefficients stabilize.

Inference with robust regression estimators. M- and MM-estimates have approximate SEs from the final weighted LS step, reasonable for large n; bootstrap SEs are usually somewhat larger and more reliable. Use a case-resampling bootstrap-t interval $(\hat{\beta}_j - \widehat{\text{se}}\, Q_{t^*}(1-\frac{\alpha}{2}), \hat{\beta}_j - \widehat{\text{se}}\, Q_{t^*}(\frac{\alpha}{2}))$ or permutation tests of $t_j = \hat{\beta}_j / \widehat{\text{se}}(\hat{\beta}_j)$ (residual shuffling for one coefficient, Section 10.2). Repeated points in bootstrap samples can upset pairwise-slope estimators (ties in x); permutation tests avoid this.

10.4 Selected exercises

Motivation. An exact permutation test for any slope, the Frisch–Waugh–Lovell identity, and the Huber location estimator.

Exercise 10.1 (Permutation test of $\beta = \beta_0$)

In $y_i = \alpha + \beta x_i + \epsilon_i$ with fixed x_i and i.i.d. errors, describe a permutation test of $H_0 : \beta = \beta_0$ against $\beta \neq \beta_0$ for a given $\beta_0 \neq 0$, and show that it is exact.

Solution.

Step 1. (Why not shuffle y.) Under H_0, $Ey_i = \alpha + \beta_0 x_i$ varies with i, so the y_i are not exchangeable.

Step 2. (Transform.) $z_i = y_i - \beta_0 x_i = \alpha + (\beta - \beta_0) x_i + \epsilon_i$; under H_0, $z_i = \alpha + \epsilon_i$ are i.i.d. and unrelated to x_i.

Step 3. (Test.) Use $b(z) = S_{xz}/S_{xx} = \hat{\beta} - \beta_0$ (or its t-statistic) over the $n!$ (or M random) permutations of z: $p = \#\{\pi : |b(z_\pi)| \geq |b(z)|\}/n!$.

Step 4. (Exactness.) z is exchangeable under H_0 and permutations form a group, so Theorem 3.3 gives $P_{H_0}(p \leq \alpha) \leq \alpha$, with equality when $\alpha\, n!$ is an integer and there are no ties; for M random permutations use Proposition 3.4.

The non-rejected β_0 form an exact confidence set (Proposition 1.2). ■

Exercise 10.4 (Partial regression (Frisch–Waugh–Lovell))

Let $X = [X_0, X_1]$ have full column rank $p+1$, with q columns in X_1. Let $M_0 = I - H_0$, $\tilde{X}_1 = M_0 X_1$ (each column of X_1 replaced by its residuals on X_0) and $\mathbf{y}_{|0} = M_0\mathbf{y}$. Consider the partial regression $\mathbf{y}_{|0} = \tilde{X}_1\boldsymbol{\gamma} + \text{error}$. Show:
(a) its least squares estimator is $\hat{\boldsymbol{\gamma}} = \hat{\boldsymbol{\beta}}_1$, and its residual vector equals the full-model residual vector $\mathbf{e}$;
(b) the F-statistic for $\boldsymbol{\gamma} = 0$, computed with $n - p - 1$ residual degrees of freedom, equals (10.1).

Solution.

(a)

Step 1. The full fit is $\mathbf{y} = X_0\hat{\boldsymbol{\beta}}_0 + X_1\hat{\boldsymbol{\beta}}_1 + \mathbf{e}$ with $X_0'\mathbf{e} = X_1'\mathbf{e} = 0$.

Step 2. Apply M_0: $M_0X_0 = 0$ and $H_0\mathbf{e} = X_0(X_0'X_0)^{-1}X_0'\mathbf{e} = 0$, so $M_0\mathbf{e} = \mathbf{e}$ and

$$\mathbf{y}_{|0} = \tilde{X}_1\hat{\boldsymbol{\beta}}_1 + \mathbf{e}.$$

Step 3. $\tilde{X}_1'\mathbf{e} = X_1'M_0\mathbf{e} = X_1'\mathbf{e} = 0$, so $\boldsymbol{\gamma} = \hat{\boldsymbol{\beta}}_1$ solves the partial normal equations $\tilde{X}_1'(\mathbf{y}_{|0} - \tilde{X}_1\boldsymbol{\gamma}) = 0$.

Step 4. If $\tilde{X}_1\mathbf{c} = 0$ then $X_1\mathbf{c} = H_0X_1\mathbf{c} \in \mathrm{col}(X_0)$, so $\mathbf{c} = 0$ by full rank. Hence $\hat{\boldsymbol{\gamma}} = \hat{\boldsymbol{\beta}}_1$ uniquely, with residual $\mathbf{e}$.

(b) The partial RSS is $\mathbf{e}'\mathbf{e} = \mathbf{y}'(I-H)\mathbf{y}$; under $\boldsymbol{\gamma} = 0$ it is $\mathbf{y}'M_0\mathbf{y} = \mathbf{y}'(I - H_0)\mathbf{y}$. The difference is $\mathbf{y}'(H - H_0)\mathbf{y}$, so

$$F = \frac{\mathbf{y}'(H - H_0)\mathbf{y}/q}{\mathbf{y}'(I-H)\mathbf{y}/(n-p-1)},$$

which is (10.1). Naive software would use $n - q$ d.f.; the correct $n - p - 1$ is the dimension of $\mathrm{col}(X)^{\perp} \ni \mathbf{e}$. ■

Exercise 10.7 (The Huber location estimator)

Fix $\hat{\tau} > 0$ and $k > 0$, let $\psi(u) = \max(-k, \min(k, u))$, $u_i(\mu) = (x_i - \mu)/\hat{\tau}$ and $g(\mu) = \sum_i \psi(u_i(\mu))$.
(a) Show that the solutions of $g(\mu) = 0$ form a nonempty closed bounded interval. Show that it is a single point unless no observation lies within $k\hat{\tau}$ of the solutions.
(b) Show that every solution $\hat{\mu}$ satisfies $\hat{\mu} = \sum_i w_i x_i / \sum_i w_i$ with $w_i = \min\{1, k/|u_i(\hat{\mu})|\}$ (the fixed-point equation of iteratively reweighted least squares).
(c) Show that if $\max_i |x_i - \bar{x}| \le k\hat{\tau}$, then $\hat{\mu} = \bar{x}$.

Solution [Supplement].

(a)

Step 1. Each $\psi(u_i(\mu))$ is continuous and nonincreasing in μ, so is g; $g(\mu) = nk$ for $\mu \le \min_i x_i - k\hat{\tau}$ and $-nk$ for $\mu \ge \max_i x_i + k\hat{\tau}$.

Step 2. By the IVT g has a root; $g^{-1}(\{0\})$ is closed (continuity), bounded (Step 1), and an interval since $g(\mu_1) = g(\mu_2) = 0$ forces $g = 0$ on $[\mu_1, \mu_2]$ by monotonicity.

Step 3. If $g \equiv 0$ on $[\mu_1, \mu_2]$, $\mu_1 < \mu_2$, every nonincreasing term with constant sum is constant there; but a term with $|u_i(\mu)| < k$ at some $\mu \in (\mu_1, \mu_2)$ is strictly decreasing near μ. So a nondegenerate root interval needs every observation at distance $\ge k\hat{\tau}$ from (μ_1, μ_2) (equally many on each side); otherwise the root is unique.

(b) For $u_i \ne 0$, $\psi(u_i) = w_i u_i$ with $w_i = \min\{1, k/|u_i|\} \in (0, 1]$; for $u_i = 0$ take $w_i = 1$. Then $0 = \sum_i w_i(x_i - \hat{\mu})/\hat{\tau}$ and $\sum_i w_i > 0$ give $\hat{\mu} = \sum_i w_i x_i / \sum_i w_i$; points beyond $k\hat{\tau}$ get weight $\propto 1/|u_i|$.

(c) At $\mu = \bar{x}$ all $|u_i| \le k$, so $g(\bar{x}) = \sum_i (x_i - \bar{x})/\hat{\tau} = 0$. Suppose the root set were $[\mu_1, \mu_2] \ni \bar{x}$, $\mu_1 < \mu_2$. If all $x_i = \bar{x}$, $\mu \in (\mu_1, \mu_2)$ near $\bar{x}$ has $|u_i(\mu)| < k$; otherwise moving μ from $\bar{x}$ toward an observation on one side gives $|x_i - \mu| < |x_i - \bar{x}| \le k\hat{\tau}$. Both contradict Step 3, so $\hat{\mu} = \bar{x}$: Huber equals the mean unless some residual exceeds $k\hat{\tau}$. ■